

Data Preparation Report

Project: retail-analytics-eda

This report documents all data type conversions, data corrections, and structural changes made during the data preparation phase (file: 03_data_preparation.sql).

1. Summary of Changes

Table	Column	Change	Notes
sales_orders	order_date	nvarchar(50) to date	598 rows with invalid date removed first
sales_orders	product_id	smallint to int	No issues found
sales_orders	unit_price	nvarchar to decimal(10,2)	No issues found
sales_orders	discount_pct	nvarchar to decimal(10,2)	1039 rows with '%' sign cleaned first
products_mmmgmeum	base_price	nvarchar(50) to decimal(10,2)	No issues found
products_mmmgmeum	launch_date	nvarchar(50) to date	4 rows with invalid date removed first
inventory_mmmgkubv	(all columns)	No changes	stock_quantity checked — max value 567, smallint sufficient

2. sales_orders: Detailed Changes

2.1 order_date: nvarchar(50) to date

When attempting to convert order_date to date type, an error appeared indicating some values could not be converted. A check was performed to identify the invalid values:

```
SELECT DISTINCT order_date
FROM sales_orders
WHERE TRY_CAST(order_date AS date) IS NULL
AND order_date IS NOT NULL;-- A message appeared indicating that some values could not be converted,
-- so we check for any invalid values.
-- It turned out that there is a date 2024-13-40. This month and day do not exist
-- and need to be analyzed to decide what to do with this value.
```

Result: invalid date '2024-13-40' found (month 13 and day 40 do not exist). Before deleting, the rows were investigated to check for patterns or links to other tables:

```
SELECT *
FROM [sales_orders (1)]
WHERE order_date = '2024-13-40';
```

	order_id	order_date	customer_id	country	product_id	quantity	unit_price	discount_pct	status
1	56173	2024-13-40	16676	SLOVAKIA	251	2	259.74	14.63	completed
2	56441	2024-13-40	53166	poland	976	2	184.57		DONE
3	56538	2024-13-40	55580	IT	352	4	158.59	24.14	SHIP
4	56970	2024-13-40	48550	SE	2190	2	56.58		CANCELLED
5	58133	2024-13-40	29405	DE	896	3	217.05	11.91	completed
6	58228	2024-13-40	47841	France	53	3	179.37	22.41	Completed
7	58483	2024-13-40	43957	slovak	2361	3	436.92	7.12	CANCELLED
8	59383	2024-13-40	57972	austria	322	2	281.43	2.66	Completed
9	59839	2024-13-40	9130	Deutschland	827	2	209.03	11.21	SHIP
10	60187	2024-13-40	3794	sweden	518	6	159.39	21.02	completed
11	60271	2024-13-40	12885	Italy	2091	3	126.02	5.61	COMPLETED
12	60500	2024-13-40	55582	spain	1685	4	279.57	0.0	COMPLETED
13	61287	2024-13-40	47314	Czechia	1821	2	161.93		complete
14	61888	2024-13-40	35847	IT	640	3	135.56	20.38	shipped
15	63424	2024-13-40	734	France	835	2	156.07	8.79	shipped
16	63682	2024-13-40	52984	IT	1104	3	226.15	0.0	done
17	63796	2024-13-40	55571	FR	1844	5	144.61	14.68	completed
18	64459	2024-13-40	986	poland	1304	6	153.64		Shipped
19	64981	2024-13-40	10736	netherlands	1132	2	281.76		done
20	65295	2024-13-40	55933	SE	269	3	358.86	25.09	Completed
21	65970	2024-13-40	46076	SE	722	2	90.97	10.27	DONE
22	66377	2024-13-40	55785	austria	508	0	71.03	0.27	Completed
23	66554	2024-13-40	29119	spain	1141	3	116.76	9.57	complete

Investigation showed no systematic pattern. Rows represent random countries, products and order values. Data loss considered acceptable (0.23% of total rows).

```
15
16
17 DELETE FROM sales_orders
18 WHERE order_date = '2024-13-40';
```

Progress	Messages
0%	No issues found
(598 rows affected)	
Completion time: 2026-05-04T20:58:39.4457556+02:00	

After removing invalid rows, the data type conversion was performed successfully:

```
ALTER TABLE sales_orders
ALTER COLUMN order_date date;--Converting data type from nvarchar(50) to date
```

2.2 product_id: smallint to int

Converted without any errors

```
30  
31  
32 ALTER TABLE sales_orders  
33 ALTER COLUMN product_id int; --  
34  
35  
36
```

% No issues found

Messages

Commands completed successfully.

Completion time: 2026-05-07T19:27:28.2272566+02:00

2.3 unit_price: nvarchar to decimal(10,2)

No problems here, conversion went through cleanly

```
33  
34 ALTER TABLE sales_orders  
35 ALTER COLUMN unit_price decimal(10,2);  
36  
37  
38
```

110 % No issues found

Messages

Commands completed successfully.

Completion time: 2026-05-07T19:28:57.5930133+02:00

2.4 discount_pct: nvarchar to decimal(10,2)

When attempting conversion, an error appeared because some values were stored with a trailing '%' sign (e.g. '15%'), which cannot be cast to a number. The invalid values were identified first:

```
SELECT DISTINCT discount_pct FROM sales_orders
WHERE TRY_CAST(discount_pct AS decimal(10,2)) IS NULL AND discount_pct IS NOT NULL;
```

The '%' sign was removed using REPLACE, so the text could be converted to a number (1039 rows affected):

```
54
55 UPDATE sales_orders
56 SET discount_pct = REPLACE (discount_pct, '%', '')
57 WHERE discount_pct LIKE '%[%]%%';
58
```

% 5 0

messages

(1039 rows affected)

Completion time: 2026-05-07T20:01:14.9604207+02:00

After cleaning, the data type conversion was performed successfully:

```
42 ALTER TABLE sales_orders
43 ALTER COLUMN discount_pct decimal(10,2);
44
```

0% 11 0

Messages

Commands completed successfully.

Completion time: 2026-06-30T14:31:48.8542468+02:00

3. products_mmmgmeum: Detailed Changes

3.1 base_price: nvarchar(50) to decimal(10,2)

The base_price column was stored as text (nvarchar). During the earlier data quality check, no negative or zero prices were found and the value range was valid (min 31.23, max 409.97). The conversion to decimal(10,2) was straightforward and completed without errors:

```
66
67 ALTER TABLE products_mmmgmeum
68 ALTER COLUMN base_price decimal(10,2);
69
70
71
```

110 % 6 0

Messages

Commands completed successfully.

Completion time: 2026-05-12T19:45:12.2063773+02:00

3.2 launch_date: nvarchar(50) to date

When attempting to convert launch_date to date type, the conversion failed with error Msg 241 ("Conversion failed when converting date and/or time from character string"), meaning at least one value could not be interpreted as a valid date:

```
71
72 ALTER TABLE products_mmmgmeum
73 ALTER COLUMN launch_date date;
74
75
76
```

110 % 5 0

Messages

Msg 241, Level 16, State 1, Line 72
Conversion failed when converting date and/or time from character string.
The statement has been terminated.

Completion time: 2026-05-12T19:47:03.8138798+02:00

I used TRY_CAST here since it returns NULL instead of throwing an error, which made it easy to spot.

```
75
76 SELECT DISTINCT launch_date
77 FROM products_mmmgmeum
78 WHERE TRY_CAST (launch_date AS date) IS NULL
79 AND launch_date IS NOT NULL;
80
81
```

110 % 5 0

Results Messages

	launch_date
1	2024-13-40

Result: the invalid date '2024-13-40' was found (month 13 and day 40 do not exist). This is the same impossible date that had already been identified and removed in the sales_orders table (see section 2.1).

Before deleting, the affected rows were investigated to check for any pattern or link to other data:

```
81 SELECT *
82 FROM products_mmmgmeum
83 WHERE launch_date = '2024-13-40';
84
```

110 % 5 0

Results Messages

	product_id	category	sub_category	base_price	launch_date
1	412	Men	Shirts	165.05	2024-13-40
2	1611	Shoes	Sneakers	267.62	2024-13-40
3	2212	Shoes	Sandals	223.88	2024-13-40
4	2398	Men	Activewear	114.43	2024-13-40

Four products were affected (product_id 412, 1611, 2212, 2398), spanning unrelated categories (Men/Shirts, Shoes/Sneakers, Shoes/Sandals, Men/Activewear) and a wide price range. No systematic pattern was found. These products did have related orders in sales_orders, but those orders had already been removed earlier during the sales_orders cleaning (the same '2024-13-40' deletion), so no orphaned orders remain.

The four rows were removed. Data loss is negligible (less than 0.1% of products):

```
89 DELETE FROM products_mmmgmeum
90 WHERE launch_date = '2024-13-40';
91
92
```

0 % 5 0

Messages

(4 rows affected)

Completion time: 2026-05-12T20:09:04.3837679+02:00

After removing the invalid rows, the data type conversion completed successfully:

```
72 ALTER TABLE products_mmmgmeum
73 ALTER COLUMN launch_date date;
```

110 % 5 0

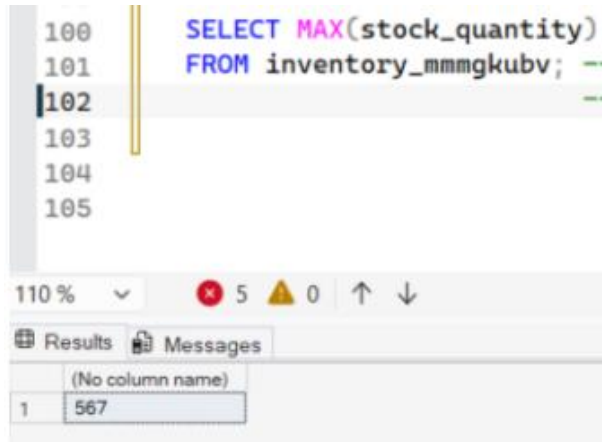
Messages

Commands completed successfully.

Completion time: 2026-05-12T20:10:01.7292580+02:00

4. inventory_mmmgkubv: No Changes

All columns were reviewed. The only candidate for a type change was stock_quantity (smallint). The maximum value was checked to decide whether a wider int type was needed:



The screenshot shows a SQL query editor with the following code:

```
100 SELECT MAX(stock_quantity)
101 FROM inventory_mmmgkubv;
102
103
104
105
```

Below the editor, the 'Results' tab is active, showing a single row of data:

(No column name)	
1	567

The maximum value was 567, well within the smallint range (32,767), so the column was left as smallint. No other changes were required:

- product_id - already int
- warehouse_country - nvarchar, correct for text data
- stock_quantity - kept as smallint (max value 567, smallint is sufficient)
- last_stock_update - already date

5. Joining Tables for Analysis

To answer the business questions, the tables had to be joined on product_id. The join logic was explored first, before building the final analytical views.

5.1 Initial join of all three tables

A first exploratory query joined all three tables together to see the combined result:

```
104 SELECT *
105 FROM sales_orders
106 INNER JOIN products_mmmgmeum ON sales_orders.product_id=products_mmmgmeum.product_id
107 INNER JOIN inventory_mmmgkubv ON sales_orders.product_id=inventory_mmmgkubv.product_id;--połączenie wszystkich tabel
108
109
110
```

order_id	order_date	customer_id	country	product_id	quantity	unit_price	discount_pct	status	product_id	category	sub_category	base_price	launch_date	product_id	warehouse_country	stock_quantity	last_stock_update
1	2023-04-25	54675	FR	2196	6	149.95	30.27	SHIPPED	2196	Accessories	Jewelry	101.59	2023-12-01	2196	PL	139	2024-01-02
2	2023-04-25	54675	FR	2196	6	149.95	30.27	SHIPPED	2196	Accessories	Jewelry	101.59	2023-12-01	2196	CZ	92	2023-10-30
3	2022-06-20	10125	AT	268	6	144.87	0.00	COMPLETED	268	Men	T-Shirts	114.39	2017-07-28	268	DE	194	2023-05-13
4	2022-06-20	10125	AT	268	6	144.87	0.00	COMPLETED	268	Men	T-Shirts	114.39	2017-07-28	268	PL	151	2023-08-22
5	2020-12-11	33966	PL	964	5	191.27	0.00	DONE	964	Men	Shirts	158.39	2021-09-16	964	CZ	103	2023-11-15
6	2020-12-11	33966	PL	964	5	191.27	0.00	DONE	964	Men	Shirts	158.39	2021-09-16	964	DE	251	2023-10-23
7	2019-06-15	9665	IT	1637	2	174.30	26.53	COMPLETED	1637	Women	Lingerie	183.50	2018-10-21	1637	CZ	277	2023-06-06
8	2019-06-15	9665	IT	1637	2	174.30	26.53	COMPLETED	1637	Women	Lingerie	183.50	2018-10-21	1637	DE	313	2023-07-10
9	2022-06-19	21017	BK	589	3	159.52	19.85	DONE	589	Accessories	Belts	152.75	2015-04-24	589	PL	0	2023-11-04
10	2022-06-18	17359	AT	1473	2	101.93	0.06	SHIPPED	1473	Kids	Top	94.98	2022-05-12	1473	CZ	364	2023-05-28
11	2022-06-18	17359	AT	1473	2	101.93	0.06	SHIPPED	1473	Kids	Top	94.98	2022-05-12	1473	PL	457	2024-04-21
12	2016-06-08	10151	FR	800	7	99.10	1.50	SHIPPED	800	Kids	Shoes	97.58	2022-12-18	800	DE	126	2024-12-18
13	2016-06-08	10151	FR	800	7	99.10	1.50	SHIPPED	800	Kids	Shoes	97.58	2022-12-18	800	PL	160	2023-11-26
14	2022-12-13	47515	ES	896	3	270.30	0.00	COMPLETED	896	Shoes	Sandals	262.03	2016-02-07	896	CZ	552	2024-03-14
15	2015-11-05	36633	FR	2329	2	147.97	7.33	DONE	2329	Women	Lingerie	169.33	2024-08-04	2329	CZ	127	2023-06-15
16	2020-01-03	39728	DE	744	2	156.56	17.73	DONE	744	Men	Activewear	105.52	2024-06-10	744	PL	323	2024-08-12
17	2020-01-03	39728	DE	744	2	156.56	17.73	DONE	744	Men	Activewear	105.52	2024-06-10	744	CZ	27	2024-12-03
18	2021-07-07	18135	CZ	1272	4	237.43	10.03	COMPLETED	1272	Women	Top	200.02	2015-08-20	1272	CZ	133	2024-06-02
19	2021-07-07	18135	CZ	1272	4	237.43	10.03	COMPLETED	1272	Women	Top	200.02	2015-08-20	1272	PL	361	2024-03-08
20	2019-09-10	38748	IT	1031	4	47.12	23.63	DONE	1031	Kids	Bottoms	41.91	2020-12-11	1031	DE	107	2024-01-04
21	2019-09-10	38748	IT	1031	4	47.12	23.63	DONE	1031	Kids	Bottoms	41.91	2020-12-11	1031	PL	106	2024-11-15
22	2017-06-07	975	ES	1898	2	118.69	4.99	COMPLETED	1898	Accessories	Belts	105.46	2015-08-03	1898	CZ	157	2024-08-09
23	2017-06-07	975	ES	1898	2	118.69	4.99	COMPLETED	1898	Accessories	Belts	105.46	2015-08-03	1898	PL	191	2024-08-15

Observation: orders were duplicated because the same product is stored in more than one warehouse. Each product_id in inventory_mmmgkubv can match several warehouse rows, so a single order line fans out into multiple rows (e.g. order_id 1 appears for both the PL and CZ warehouse). This means a sales_orders + inventory join cannot reliably attribute an order to one specific warehouse, which limits the depth of inventory and logistics analysis (business questions Q3 and Q6).

5.2 Recommended data collection improvement

The most impactful future improvement would be to add a shipped_from_warehouse column to the sales_orders table, populated automatically when an order status changes to SHIPPED or COMPLETED, recording which warehouse fulfilled the order. This would enable:

- Accurate inventory analysis - knowing which warehouse stock was used for each order
- Overstocking detection by country - comparing actual shipments against stock levels per warehouse
- Logistics optimization and more actionable insights for management

To work around the limitation above, two focused VIEWS were created, each joining sales_orders with only one other table.

5.3 View for product analysis: seasonality_VS_margin

This view supports the product-related questions, starting with business question Q4 ("Does seasonality affect the margin?"). The sales_orders + products_mmmgmeum join was tested first, then saved as a view:

```
115 CREATE VIEW seasonality_VS_margin AS
116 SELECT sales_orders.product_id, order_date, quantity, unit_price, base_price, status
117 FROM sales_orders
118 INNER JOIN products_mmmgmeum ON sales_orders.product_id=products_mmmgmeum.product_id;
119
```

110 % 6 0

Messages

Commands completed successfully.

Completion time: 2026-05-20T20:15:54.1704212+02:00

```
120 SELECT TOP 10 *
121 FROM seasonality_VS_margin;
122
123
124
125
```

110 % 6 0

Results Messages

	product_id	order_date	quantity	unit_price	base_price	status
1	2196	2023-04-25	6	149.95	101.59	SHIPPED
2	268	2022-06-20	6	144.87	114.39	COMPLETED
3	964	2020-12-11	5	191.27	158.39	DONE
4	1637	2019-06-15	2	174.30	183.50	COMPLETED
5	589	2022-08-19	3	159.52	152.75	DONE
6	1473	2022-06-18	2	101.93	94.98	SHIPPED
7	800	2016-06-08	7	99.10	97.58	SHIPPED
8	896	2022-12-13	3	270.30	262.03	COMPLETED
9	2329	2015-11-05	2	147.97	169.33	DONE
10	744	2020-01-03	2	156.56	105.52	DONE

The view was then extended with ALTER VIEW: the discount_pct column was added because it was needed for business question Q5 ("Do promotions have a negative impact on profit?"), and a computed revenue column (unit_price * quantity) was added for the revenue analysis. Final definition:

```
123 ALTER VIEW seasonality_VS_margin AS
124 SELECT sales_orders.product_id, order_date, quantity, unit_price, base_price, discount_pct, status
125 FROM sales_orders
126 INNER JOIN products_mmmgmeum ON sales_orders.product_id=products_mmmgmeum.product_id; -- dodanie kolumny discount_pct dal analizy pytania nr. 5
127
128
```

110 % 7 0

Ln: 126

Messages

Commands completed successfully.

Completion time: 2026-05-20T20:02:02.1447545+02:00

5.4 View for inventory analysis: sales_inventory

This view supports business questions Q3 ("Do we have an inventory problem?") and Q6 ("Which countries are overstocked?"). The sales_orders + inventory_mmmgkubv join was checked first, then saved as a view:

```
132 SELECT sales_orders.product_id, order_date, quantity, country, warehouse_country, stock_quantity, last_stock_update
133 FROM sales_orders
134 INNER JOIN inventory_mmmgkubv ON sales_orders.product_id=inventory_mmmgkubv.product_id
135
```

	product_id	order_date	quantity	country	warehouse_country	stock_quantity	last_stock_update
1	2196	2023-04-25	6	FR	PL	139	2024-01-02
2	2196	2023-04-25	6	FR	CZ	92	2023-10-30
3	268	2022-06-20	6	AT	DE	194	2023-05-13
4	268	2022-06-20	6	AT	PL	151	2023-08-22
5	964	2020-12-11	5	PL	CZ	103	2023-11-15
6	964	2020-12-11	5	PL	DE	251	2023-10-23
7	1637	2019-06-15	2	IT	CZ	277	2023-06-06
8	1637	2019-06-15	2	IT	DE	313	2023-07-10
9	589	2022-08-19	3	SK	PL	0	2023-11-04
10	1473	2022-06-18	2	AT	CZ	364	2023-05-28
11	1473	2022-06-18	2	AT	PL	457	2024-04-21
12	800	2016-06-08	7	FR	DE	126	2024-12-18
13	800	2016-06-08	7	FR	PL	160	2023-11-26
14	896	2022-12-13	3	ES	CZ	552	2024-03-14
15	2329	2015-11-05	2	FR	CZ	127	2023-06-15
16	744	2020-01-03	2	DE	PL	323	2024-08-12
17	744	2020-01-03	2	DE	CZ	27	2024-12-03
18	1272	2021-07-07	4	CZ	CZ	133	2024-06-02
19	1272	2021-07-07	4	CZ	PL	361	2024-03-08
20	1308	2023-04-09	4	NL	CZ	3	2024-09-20
21	1031	2019-09-10	4	IT	DE	107	2024-01-04

```
135
136 CREATE VIEW sales_inventory AS
137 SELECT sales_orders.product_id, order_date, quantity, country, warehouse_country, stock_quantity, last_stock_update
138 FROM sales_orders
139 INNER JOIN inventory_mmmgkubv ON sales_orders.product_id=inventory_mmmgkubv.product_id
140
141
142
```

Messages

Commands completed successfully.

Completion time: 2026-05-20T20:30:58.6956083+02:00

Preview of the raw join, tested before creating the view

Verified with a preview (note: subject to the warehouse duplication limitation described in 5.1 the same product appears once per warehouse):

```
146 SELECT TOP 10 *
147 FROM sales_inventory;
148
149
```

	product_id	order_date	quantity	country	warehouse_country	stock_quantity	last_stock_update
1	2196	2023-04-25	6	FR	PL	139	2024-01-02
2	2196	2023-04-25	6	FR	CZ	92	2023-10-30
3	268	2022-06-20	6	AT	DE	194	2023-05-13
4	268	2022-06-20	6	AT	PL	151	2023-08-22
5	964	2020-12-11	5	PL	CZ	103	2023-11-15
6	964	2020-12-11	5	PL	DE	251	2023-10-23
7	1637	2019-06-15	2	IT	CZ	277	2023-06-06
8	1637	2019-06-15	2	IT	DE	313	2023-07-10
9	589	2022-08-19	3	SK	PL	0	2023-11-04
10	1473	2022-06-18	2	AT	CZ	364	2023-05-28

6. Analytical Queries Built on the Views (SQL)

The views above were then used to run the analytical SQL for the business questions. The key SQL-level findings are summarised below; they are developed further, with visualisations, in the analysis report (Power BI).

6.1 Q1: Are prices really growing? (YoY)

Average revenue per year was calculated from seasonality_VS_margin, grouped by YEAR(order_date).

```
150 SELECT YEAR(order_date) AS year, CAST(ROUND(AVG(revenue),2)AS decimal(10,2)) AS average
151 FROM seasonality_VS_margin
152 GROUP BY YEAR(order_date)
153 ORDER BY YEAR(order_date) ASC;
```

	year	average
1	2015	1042.64
2	2016	1103.24
3	2017	1185.15
4	2018	1197.38
5	2019	1273.97
6	2020	1293.59
7	2021	1328.24
8	2022	1303.74
9	2023	1408.88
10	2024	1501.44

Finding: average revenue grows year over year, from roughly 1042 in 2015 to 1501 in 2024, with one exception a slight drop in 2022 compared to 2021.

6.2 Q2: Is revenue growth driven by volume or price?

Yearly total quantity and average unit price were compared, also from seasonality_VS_margin.

Finding: revenue growth is driven mainly by rising prices, not volume. Quantities stay fairly stable (roughly 161k to 174k per year), while the average price increases steadily from 155 in 2015 to 212 in 2024.

```
157 SELECT YEAR(order_date) AS year,SUM(quantity) AS SUM_QUANTITY, CAST(ROUND(AVG(unit_price),2)AS decimal(10,2)) AS AVG_UNIT_PRICE
158 FROM seasonality_VS_margin
159 GROUP BY YEAR(order_date)
160 ORDER BY YEAR(order_date) ASC;
161
```

	year	SUM_QUANTITY	AVG_UNIT_PRICE
1	2015	163523	155.83
2	2016	166402	163.25
3	2017	173310	168.74
4	2018	161182	174.84
5	2019	171627	181.77
6	2020	168048	188.19
7	2021	165823	193.72
8	2022	162859	199.60
9	2023	169116	206.82
10	2024	174890	212.11

6.3 Q3: Do we have an inventory problem?

Quantities sold were compared with stock levels using sales_inventory, filtered to 2024.

```
164 SELECT product_id, order_date, SUM(quantity) AS SUM_quantity_product, SUM(stock_quantity) AS SUM_STOCK, last_stock_update
165 FROM sales_inventory
166 WHERE YEAR(order_date) IN (2024, 2023)
167 GROUP BY product_id, order_date, last_stock_update
168 ORDER BY product_id ASC;
169
```

	product_id	order_date	SUM_quantity_product	SUM_STOCK	last_stock_update
1	1	2023-02-07	1	430	2024-10-11
2	1	2023-05-04	1	430	2024-10-11
3	1	2023-07-17	3	129	2024-09-17
4	1	2023-04-04	2	430	2024-10-11
5	1	2023-05-31	1	430	2024-10-11
6	1	2023-01-11	3	129	2024-09-17
7	1	2023-01-11	3	430	2024-10-11
8	1	2023-02-20	3	129	2024-09-17
9	1	2023-04-04	2	129	2024-09-17
10	1	2023-12-17	2	129	2024-09-17
11	1	2023-12-17	2	430	2024-10-11
12	1	2023-05-04	1	129	2024-09-17
13	1	2023-07-29	3	129	2024-09-17
14	1	2023-02-20	3	430	2024-10-11
15	1	2023-02-07	1	129	2024-09-17
16	1	2023-07-29	3	430	2024-10-11
17	1	2023-07-12	2	129	2024-09-17
18	1	2024-06-14	4	129	2024-09-17
19	1	2024-01-10	3	129	2024-09-17
20	1	2024-07-26	5	430	2024-10-11
21	1	2024-05-26	3	129	2024-09-17
22	1	2023-07-12	2	430	2024-10-11
23	1	2023-05-31	1	129	2024-09-17

Finding: a fully reliable inventory analysis is not possible, because last_stock_update does not align with order_date (e.g. an order from 2024 but a stock update from 2023, or an order from 2022 with inventory levels from 2024). This makes it difficult to draw meaningful conclusions and is consistent with the warehouse limitation described in section 5.1.

6.4 Q4: Seasonality of sales

Quantity sold was aggregated by year and month (CONCAT of year and month) from seasonality_VS_margin to reveal seasonal patterns across all products.

```
177 SELECT CONCAT(YEAR(order_date), '-', MONTH(order_date)) AS year_month, AVG(discount_pct) AS AVG_DISCOUNT, SUM(revenue) AS SUM_REVENUE, SUM(quantity) AS SUM_QUANTITY
178 FROM seasonality_VS_margin
179 GROUP BY YEAR(order_date), MONTH(order_date)
180 ORDER BY YEAR(order_date), MONTH(order_date)
```

	year_month	AVG_DISCOUNT	SUM_REVENUE	SUM_QUANTITY
1	2015-1	11.051000	1042110.00	12301
2	2015-2	11.077774	807900.34	8803
3	2015-3	10.050684	2122903.89	13302
4	2015-4	11.030377	1378868.71	9439
5	2015-5	10.860025	2067056.90	12198
6	2015-6	11.196761	2180818.48	14302
7	2015-7	11.341487	2312280.04	16144
8	2015-8	11.040156	1736659.32	11510
9	2015-9	10.998821	1995694.70	13214
10	2015-10	11.012388	2012425.91	13424
11	2015-11	10.010213	2790348.57	17418
12	2015-12	11.250084	3878754.38	24325
13	2016-1	11.142088	1050706.47	8629
14	2016-2	11.107122	1287963.02	6232
15	2016-3	11.215842	2394838.07	14030
16	2016-4	11.038108	1821480.18	10618
17	2016-5	10.874874	2002623.84	11733
18	2016-6	10.802133	2142904.87	11627
19	2016-7	11.022649	2487580.71	16086
20	2016-8	10.971511	2058413.17	13708
21	2016-9	11.254380	2200147.07	13881
22	2016-10	10.719304	2181780.80	13625
23	2016-11	10.823888	3247562.43	18732

6.5 Q5: Do promotions have a negative impact on profit?

Average discount, total revenue and total quantity were aggregated by month from seasonality_VS_margin to look for a relationship between discounts and sales.

Finding: based on this data, the correlation between discount and sales is not clear. A Pearson correlation of -0.01 confirms effectively no linear relationship.

```
183 SELECT CONCAT(YEAR(order_date), '-', MONTH(order_date)) AS year_month, CAST(ROUND(AVG(discount_pct), 2) AS decimal(10,2)) AS AVG_DISCOUNT, SUM(revenue) AS SUM_REVENUE, SUM(quantity) AS SUM_QUANTITY
184 FROM seasonality_VS_margin
185 GROUP BY YEAR(order_date), MONTH(order_date)
187 ORDER BY YEAR(order_date), MONTH(order_date); -- The correlation between discount and sales is not clear based on this data.
```

	year_month	AVG_DISCOUNT	SUM_REVENUE	SUM_QUANTITY
1	2015-1	11.05	1942018.98	12301
2	2015-2	11.08	937589.34	6803
3	2015-3	10.86	2123583.89	13302
4	2015-4	11.54	1370668.71	9424
5	2015-5	10.87	2007256.95	12356
6	2015-6	11.16	2150515.48	14302
7	2015-7	11.34	2312260.54	15144
8	2015-8	11.50	1734669.32	11510
9	2015-9	11.00	1999594.78	13214
10	2015-10	11.01	2012425.91	13424
11	2015-11	10.81	2790346.37	17418
12	2015-12	11.36	3976754.39	24325
13	2016-1	11.14	1550186.47	9629
14	2016-2	11.16	1267563.02	8232
15	2016-3	11.22	2394636.07	14030
16	2016-4	11.04	1621400.16	10619
17	2016-5	10.87	2002623.64	11733
18	2016-6	10.88	2142594.67	11637
19	2016-7	11.02	2457880.71	16088
20	2016-8	10.97	2058413.17	13708

```
188
189 SELECT
190     (AVG(disc * rev) - AVG(disc) * AVG(rev)) /
191     (STDEVP(disc) * STDEVP(rev)) AS correlation
192 FROM (
193     SELECT
194         CONCAT(YEAR(order_date), '-', MONTH(order_date)) AS ym,
195         AVG(discount_pct) AS disc,
196         SUM(revenue) AS rev
197     FROM seasonality_VS_margin
198     GROUP BY YEAR(order_date), MONTH(order_date)
199 ) correlation; --correlation between discount and revenue
200
201
202
```

	correlation
1	-0.0120104748149217

Note: a margin-level analysis (relevant to Q4) is limited by inconsistencies between unit_price, base_price and discount_pct identified earlier, so margin figures should be treated with caution. The findings above form the SQL basis for the analysis report, where they are expanded with Power BI visualisations.